

$$\bar{R} = 30.0\%/10 = 3.0\%.$$

2. The geometric mean return over the 10 years is *closest* to:

- A. 2.94 percent.
- B. 2.97 percent.
- C. 3.00 percent.

Solution:

B is correct. The geometric mean return is calculated as follows:

$$\bar{R}_G = \sqrt[10]{(1 + 0.045) \times (1 + 0.06) \times \dots \times (1 + 0.055) \times (1 + 0.04)} - 1$$

$$\bar{R}_G = \sqrt[10]{1.3402338} - 1 = 2.9717\%.$$

MONEY-WEIGHTED AND TIME-WEIGHTED RETURN

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compare the money-weighted and time-weighted rates of return and evaluate the performance of portfolios based on these measures

The arithmetic and geometric return computations do not account for the timing of cash flows into and out of a portfolio. For example, suppose an investor experiences the returns shown in Exhibit 2. Instead of only investing EUR1.0 at the start (Year 0) as was the case in Exhibit 2, suppose the investor had invested EUR10,000 at the start, EUR1,000 in Year 1, and EUR1,000 in Year 2. In that case, the return of –50 percent in Year 1 significantly hurts her given the relatively large investment at the start. Conversely, if she had invested only EUR100 at the start, the absolute effect of the –50 percent return on the total return is drastically reduced.

Calculating the Money Weighted Return

The **money-weighted return** accounts for the money invested and provides the investor with information on the actual return she earns on her investment. The money-weighted return and its calculation are similar to the internal rate of return and a bond's yield to maturity. Amounts invested are cash outflows from the investor's perspective and amounts returned or withdrawn by the investor, or the money that remains at the end of an investment cycle, is a cash inflow for the investor.

For example, assume that an investor invests EUR100 in a mutual fund at the beginning of the first year, adds another EUR950 at the beginning of the second year, and withdraws EUR350 at the end of the second year. The cash flows are presented in Exhibit 10.